

Layered Threat-Timeline-Driven Quantum-Resistant Federated Learning (QRFL) Architecture

Forecast → PQC agility → healthcare FL → Fabric governance → Mosca migration policy

L1 Forecast

Quantum Threat Timeline & Forecasting Engine

Hardware Corpus (n=20)
IBM, Google, Rigetti,
Quantinuum, Atom Computing
Status-classified, 2016–2026

Forecasting Models
Log-linear OLS ($T_d=1.86$ y)
Logistic saturation ($L\approx 19.26M$)
ECDSA-256 crossing ≈ 2050.19

Risk Trigger & UQ
Bootstrap 95% lower bound 2042.08
Feeds Mosca horizon Z
Planning heuristic, not a compromise date

Cryptographic Parameter Selection
(NIST Level 3)

L2 PQC

Post-Quantum Cryptographic & Crypto-Agility Engine

Hybrid PKI / CA
Dual X.509 issuance
ECDSA-P256 + ML-DSA-65
Hybrid certificates in transition

KEM Session Channel
ML-KEM-768 + ephemeral ECDHE
HKDF-derived K_{session}
NIST Level 3, hybrid then native

Digital Signatures
ML-DSA-65 (node / FL updates)
SLH-DSA-SHAKE-256s (archival)
Long-lived attestation anchors

Crypto-Agility Engine
Runtime algorithm swapping
Cipher-suite negotiation
Policy-driven suite updates

Session Encryption & Mask Primitives
(ML-KEM-768)

L3 FL

Privacy-Preserving Healthcare Federated Learning Layer

Clinical Edge Nodes
Hospitals $1, \dots, K$; local EHR
PneumoniaMNIST benchmark
Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$

Lattice-Compatible Masking
Additive ring masking
 $W_u^* = W_u + M_u, \sum M_u = 0$
PQ-ready secure aggregation

Flower Aggregator & Defense
FedAvg with robust filters
Coordinate-wise Median; Krum
Byzantine / poisoning resistance

Signed Masked Updates Proposal
($W_u^*, \sigma_u, \text{Cert}_u$)

L4 Ledger

Permissioned Blockchain Governance & Ledger Layer

Consortium Network
3-peer Hyperledger Fabric v2.5
Permissioned hospital consortium
Channel qrfchannel

Ordering Engine
Raft crash-fault tolerance
etcdraft ordering service
Live test-network validated

Smart Chaincode
Contract flupdate
Validates ML-DSA-65 on W_u^*
SubmitUpdate invoke path

Ledger & DCRL
Immutable audit trail
Decentralized revocation list
Consent / provenance records

L5 Migration

Strategic Migration & Decision Support Layer

Mosca Risk Assessment
 $X + Y > Z$
 $X=15$ y PHI retention, $Y=5$ y migration
Trigger $Z=2042.08$ (bootstrap 95% LB)
Finish native PQC before 2041

Phased Roadmap

I 2026–2029
Inventory / hybrid TLS

II 2029–2032
Archival PQC

III 2032–2036
Native FL / ledger

IV 2036–2041
Native-PQC policy

Threat Horizon
($Z=2042.08$)

Governance Deadlines
& Migration Milestones

On-Chain DCRL Revocation
& Agility Policies